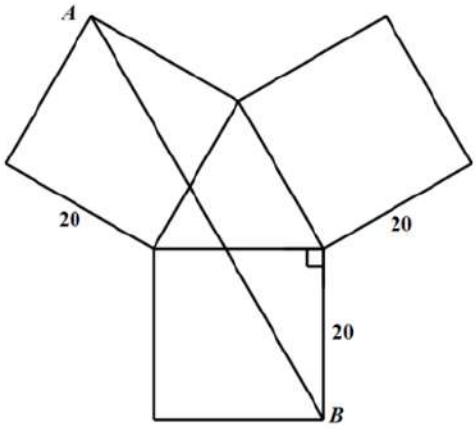
APMOPS 2025 Round1 Questions and Answers

1. The distance from A to B is represented as $a+\sqrt{b}$. Find $a+b$.



【Answer】 1220

【Solution】 As shown in the figure, we can get $\angle HAB = \angle FBA$

Extend AH and BF to meet at E, $\triangle ABE$ is an equilateral triangle.

And $\angle GEF = 60° \div 2 = 30°$.

In right triangle EFG, $\angle GEF = 30°$, so $EG = 2FG = 20$

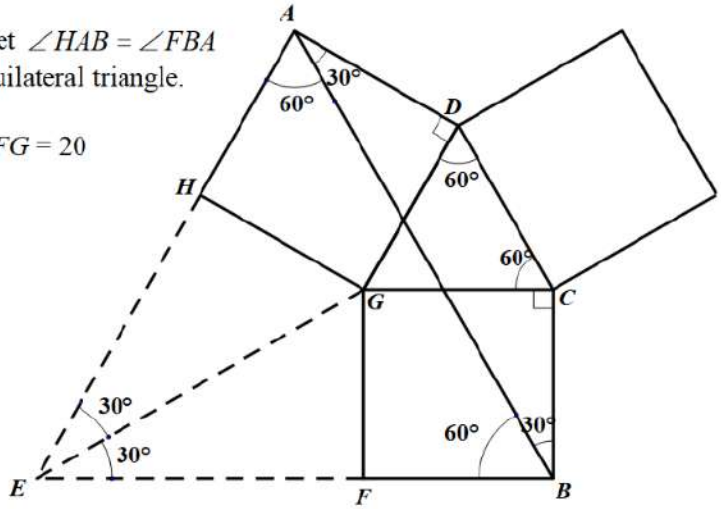
$EF^2 = EG^2 - FG^2 = 20^2 - 10^2 = 1200$

$EF = \sqrt{1200}$

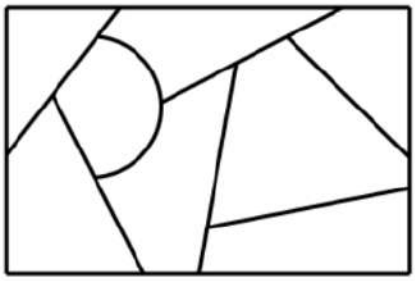
$a+\sqrt{b} = 20+\sqrt{1200}$

$a = 20, b = 1200$

So $a+b = 1220$



2. Find the number of ways to colour the above diagram using 4 different colours, given that no adjacent regions can be of the same colour.



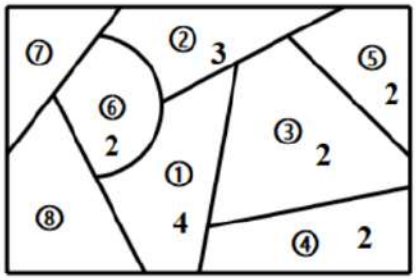
【Answer】 576

【Solution】

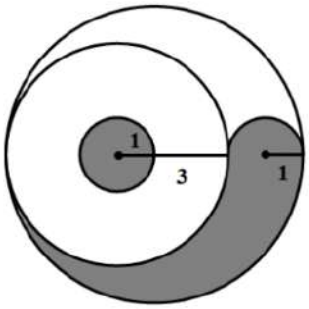
If position 1 and 7 are the same, we have $4\times3\times2\times2\times2\times2\times1\times2$

If position 1 and 7 are different, we have $4\times3\times2\times2\times2\times2\times1\times1$

Summing up, we get 576.

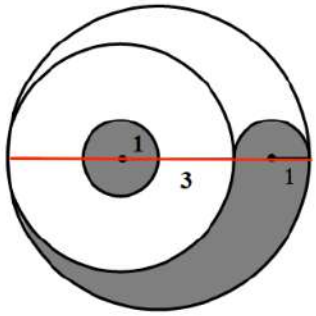


3. The radius of the figures are as follows: big circle = 4, medium circle = 3, small circle = 1 and semicircle = 1.
The ratio of the shaded to unshaded area is $m:n$. Find the value of $m+n$.



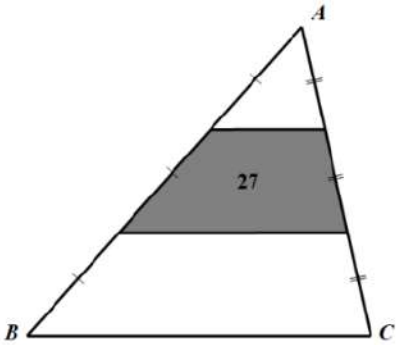
【Answer】 16

【Solution】 Area of shaded part $=\pi\times1^2+\frac{\pi\times1^2}{2}+\left(\frac{\pi\times4^2-\pi\times3^2}{2}\right)=5\pi$

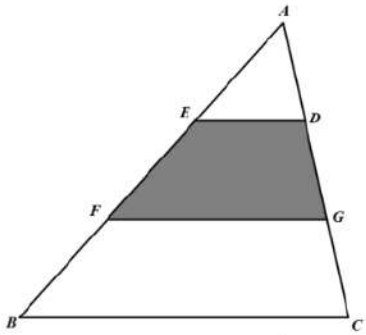


Area of unshaded part $=\pi\times4^2-5\pi=11\pi$
So the ratio is $5\pi:11\pi=5:11$
$m+n=16$

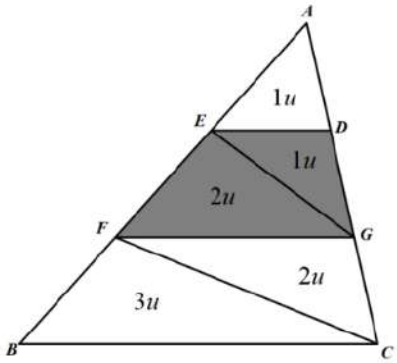
4. In the above figure, the shaded area is 27. Line AB is split into 3 equal parts and line AC is also split into 3 equal parts. Find the area ABC.



【Answer】 81
【Solution】 **Method 1:**
ΔADE is similar to ΔAGF and ΔACB.
Let the area of ΔADE be $1u$, the area of ΔAGF will be $(2\times2)u=4u$, and the area of ΔACB will be $(3\times3)u=9u$.
$4u-1u=3u=27, 9u=81$



Method 2:
Let the area of ΔADE be $1u$
Using Equal-Height Model
The area of $\Delta ADE=1u$
The area of $\Delta FEG=\Delta AEG=2u$
The area of $\Delta CFG=\Delta AFG\div2=2u$
The area of $\Delta BCF=\Delta ACF\div2=3u$
So the area of ΔACB will be $27\div3\times9=81$.



5. Tim had a certain number of marbles. He told this number to Tom, who then told John. John told Jim, Jim told Bob, and Bob told Adam.
However,
(a) When Tom told John, he overcounted 9 to the number.
(b) When John told Jim, he swapped the last two digits of the number.
(c) When Jim told Bob, he swapped the first and third digits of the number.
(d) Finally, when Bob told Adam, he doubled the number.
Surprisingly, Adam ended up with the same number of marbles as Tim had initially. How many marbles did Tim start with?

【Answer】 2052

【Solution】 Let the number of marbles that Tom told John be $\overline{abcd}$.

Tim told Tom	Tom told John	John told Jim	Jim told Bob	Bob told Adam
$\overline{abcd}-9$	$\overline{abcd}$	$\overline{abdc}$	$\overline{dbac}$	$\overline{dbac}\times 2$

So $\overline{dbac}\times 2=\overline{abcd}-9$

$2000d+200b+20a+2c=1000a+100b+10c+d-9$

$1999d+100b+9=980a+8c$

$1999d+100b+9=4(245a+2c)$

$1999d+100b+9\equiv 4(245a+2c)(\bmod 4)$

$3d+1\equiv 0(\bmod 4)$

$3d\equiv 3(\bmod 4)$

$d\equiv 1(\bmod 4)$

And $980a+8c\le 980\times 9+8\times 9=8892$, so $d\le 3$

Now $d=1$, $1999+100b+9=980a+8c\le 1999+100\times 9+9=2908$

So $a=2$, $1999+100b+9=980\times 2+8c$, $48+100b=8c$

$b=0$, $48=8c$, then $c=6$

So $\overline{abcd}=2061$

$\overline{abcd}-9=2052$

6. Find the smallest possible integer n such that any prime number squared plus n will not be a prime number.

【Answer】 5

【Solution】

Since all the prime numbers besides 2 are odd, the square of an odd number must be odd, so we only need to add an odd number to a prime number squared beside 2, then the sum must be an even, not prime number.

Now we only need to consider the prime number, 2,

$2^2=4$

$2^2+1=5$, 5 is prime

$2^2+3=7$, 7 is prime

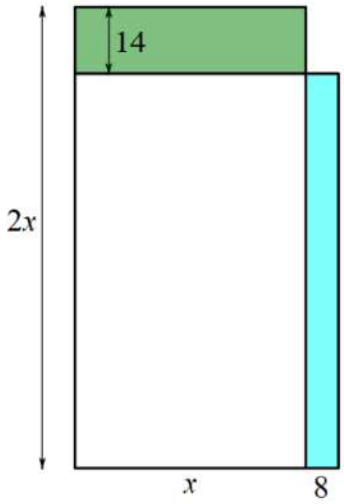
$2^2+5=9$, not prime

So the smallest possible integer n is 5.

7. 2025 is a four-digit number where 2025 equals $(20+25)^2$. It is known that there are two other four-digit numbers ABCD where $(AB+CD)^2 = ABCD$.
Find the sum of the two other four-digit numbers.

【Answer】 12826
【Solution】
$(x+y)^2 = 100x + y$
$(x+y)^2 = x + y + 99x$
$(x+y)^2 - (x+y) = 99x$
$(x+y)(x+y-1) = 99x$
$x + y = 45, 55, 99$
$55^2 + 99^2 = 12826$

8. There is a rectangle tablecloth where its length is two times the width. After shortening the length by 14cm and extending the width by 8cm, the area remains the same. Find the perimeter of the new tablecloth.



【Answer】 324
【Solution】 As shown in the figure,
Let the original width be x, then the length is $2x$
Since the area remains the same, remove the overlapped part, the area of the remaining 2 parts are the same.
$14x = 8(2x-14)$
Then $x = 56$
the perimeter of the new tablecloth: $\left[(2x-14)+(x+8)\right]\times 2 = 324$

9. The product of two positive integers is equal to 10 times the sum of the two integers and 15 times the difference between the two integers. Find the sum of the two integers.

【Answer】 72
【Solution】 $\begin{matrix} A+B=\frac{1}{10}AB \\ A-B=\frac{1}{15}AB \end{matrix} \rightarrow \begin{matrix} A=\frac{1}{2}\times(\frac{1}{10}+\frac{1}{15})AB \\ B=\frac{1}{2}\times(\frac{1}{10}-\frac{1}{15})AB \end{matrix} \rightarrow \begin{matrix} A=\frac{1}{12}AB \\ B=\frac{1}{60}AB \end{matrix} \rightarrow \begin{matrix} A=60 \\ B=12 \end{matrix}$

10. How many rectangles are there such that the sum is divisible by 5?

21	22	23	24	25
16	17	18	19	20
11	12	13	14	15
6	7	8	9	10
1	2	3	4	5

【Answer】 71

【Solution】 divisible by 5 → remainder is 0 when divided by 5

So we can change the numbers in the table to the remainder when divided by 5.

1	2	3	4	0
1	2	3	4	0
1	2	3	4	0
1	2	3	4	0
1	2	3	4	0

Rectangles in 1 row:

0				
2	3			
1	2	3	4	
1	2	3	4	0

There are 4×5 rectangles

Rectangles in 2 row:

0				
0				
2	3			
2	3			
1	2	3	4	
1	2	3	4	
1	2	3	4	0
1	2	3	4	0

There are 4×4 rectangles

Similarly,
Rectangles in 3 row: 4×3 rectangles
Rectangles in 4 row: 4×2 rectangles
Rectangles in 5 row:
The sum of any column is a multiple of 5.
So there are $5+4+3+2+1=15$ rectangles
Total: $4\times5+4\times4+4\times3+4\times2+15=71$

11. How many possible values of prime number N are there such that $(4900-N)$ is a square number?

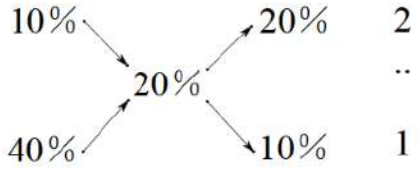
【Answer】 1
【Solution】
Let the square number $4900-N=k^2$
$N=4900-k^2$
$=70^2-k^2$
$=(70-k)(70+k)$
Since N is a prime number, which means it has only 1 and itself, 2 factors.
So $70-k=1,\ 70+k=N$.
so only 1 possible value for N.

12. If 29 February 2024 is a Thursday, when is the next year when 29 February is a Thursday?

【Answer】 2052
【Solution】 Every 4 years, $365\times3+366\equiv5(\text{mod }7)$
As we need it to be a multiple of 7, that means $7\times4=28$ years passed.
$2024+28=2052$

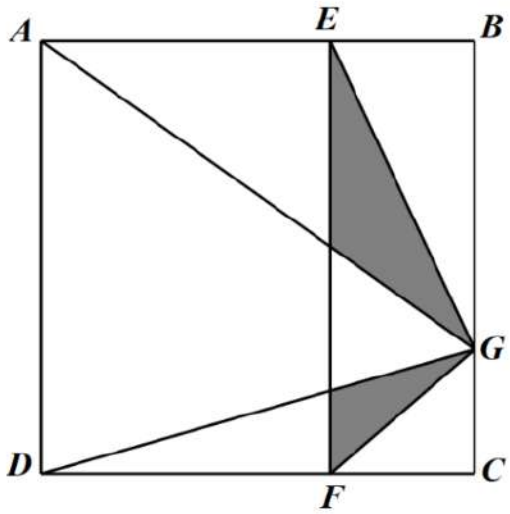
13. How many litres of 10% alcohol solution must be mixed with 40% alcohol solution to obtain 300L of 20% alcohol solution.

【Answer】 200
【Solution】 Using the cross method,



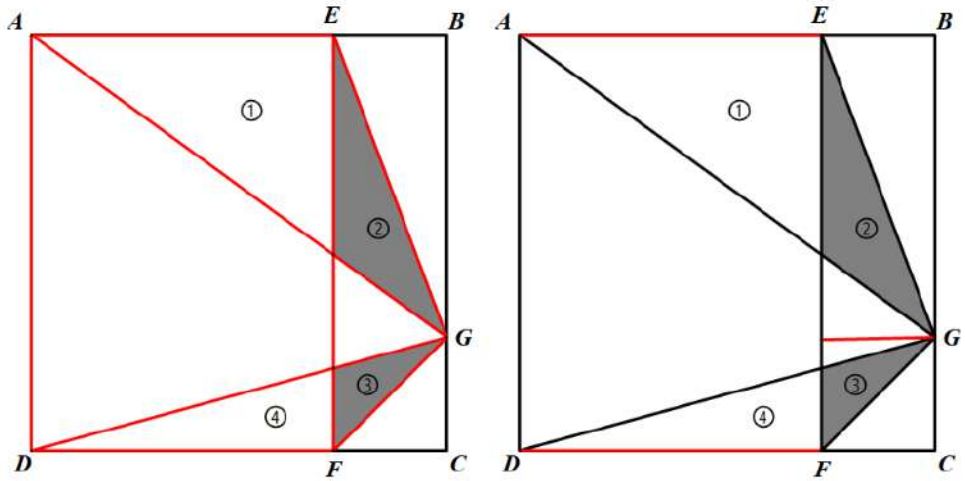
10% alcohol 【Solution】 $300\div(2+1)\times2=200$ L.

14. ABCD is a square with side length of 24. BG = 17, BE = 8, DF = 16. What is the total area of the shaded region?



【Answer】 64

【Solution】



① + ② + ③ + ④ = half of the rectangle ADFE = $24 \times 16 \div 2 = 192$

Base of ① and ② are the same, height of ① double of ②, so ① = 2②.

Area of shaded part = $192 \div 3 = 64$

15. There are 27 identical dice with 1 on the opposite of 6, 2 on the opposite of 5, and 3 on the opposite of 4. If they are formed into a $3 \times 3 \times 3$ cube, what is the maximum sum of the exposed numbers?

【Answer】 288

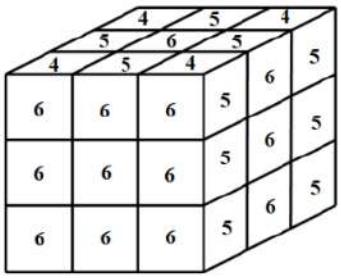
【Solution】 To get the max,

6 cubes exposing only one face (center of each face): $6 \times 6 = 36$

12 cubes exposing only two faces (middle of each edge): $(6+5) \times 12 = 132$

8 cubes exposing only three faces (each vertex): $(6+5+4) \times 8 = 120$

Total: $36 + 132 + 120 = 288$



16. A "beautiful" pair of different numbers is one that has the following properties.
(a) Both numbers are at least 25% of the other number.
(b) At least one number is between 25 to 28(inclusive).
How many "nice" pairs are there?

【Answer】 390

【Solution】 If one of the numbers is 25, then the other number ranges from 7~100. (94 numbers, but need to remove 25)

If one of the numbers is 26, then the other number ranges from 7~104. (98 numbers, but need to remove 25, 26)

If one of the numbers is 27, then the other number ranges from 7~108. (102 numbers, but need to remove 25, 26, 27)

If one of the numbers is 28, then the other number ranges from 7~112. (106 numbers, but need to remove 25, 26, 27, 28)

Hence, there are a total of $400 - 10 = 390$ pairs.

17. If $\frac{m}{3}$ and $3m$ are both 5-digit numbers, how many values are there for m ?

【Answer】 1112

【Solution】 $\begin{matrix} \frac{m}{3} \geq 10000 \\ 3m \leq 99999 \end{matrix} \rightarrow \begin{matrix} m \geq 30000 \\ m \leq 33333 \end{matrix}$

As it needs to be a multiple of 3, $(33333-30000) \div 3+1=1112$

18. How many numbers below 2025 are coprime to only one of 20 or 25?

【Answer】 810

【Solution】 $20=2^2 \times 5$, $25=5^2$, since we can only coprime to only one, we can just consider it to be coprime to 25 but not coprime to 20. That means we just need to check for multiples of 2, but not multiples of 5.

The number of multiples of 2 are: $2024 \div 2=1012$

The number of multiples of 10 are: $2024 \div 10=202R4$

Hence, there are a total of $1012-202=810$ numbers.

19. A pest control management caught 94 rats and 49 squirrels on the first day. Every day, they catch 2 more squirrels and 3 less rats than the previous day. On what day the total number of squirrels caught exceeds the total number of rats caught.

【Answer】 20

【Solution】 the first day, the rats is $94-49=45$ more than squirrels.

the second day, the rats is $91-51=40$ more than squirrels.

the third day, the rats is $88-53=35$ more than squirrels.

...

the difference between the total number of squirrels caught and the total number of rats caught:

$45+40+35+\cdots+5+0-5-10-\cdots$

when the total number of squirrels caught = the total number of rats caught:

$45+40+35+\cdots+5+0-5-10-\cdots-40-45$

It need $45 \times 2 \div 5+1=19$ days. So the 20th day the total number of squirrels caught exceeds the total number of rats caught.

20. How many 4-digit numbers are there that have 1 pair of same digits exactly? (e.g. 7174 and 6008 are counted while 2002 and 1011 are not)

【Answer】 3888

【Solution】 As we have two same digits, we can let the number be $AABC$, where each alphabet represents a different number.

Firstly, for the two digits that are the same, there are a total of 6 different ways to place them.

Then we think of how many ways can we select those numbers. As the number cannot start from 0, we only have 9 ways(1~9) to select the first digit, followed by 9 ways to select the second, and 8 ways to select the last.

As such, we have a total of $6 \times 9 \times 9 \times 8=3888$ ways

21. We define 1 as a special number. For any number greater than 1, it is a special number only if it is coprime with the sum of all smaller special numbers. 2 is special because it is coprime with 1. 3 is not special because it is not coprime with the sum of smaller special numbers. 4 is special because it is coprime with the sum $1+2+3=6$.
Find the 20th special number.

【Answer】 29
【Solution】 The numbers are in the pattern of 1, 2, 4, 5, 7, 8, 10, 11…
We realize that multiples of 3 are excluded.
Hence, that means the 20th number will be $20 \div 2 \times 3 = 30$, but we must subtract one as every group there are only the first two numbers. $30-1=29$

22. Tom takes 21 minutes to run from his house to Jerry's house, while Jerry takes 28 minutes to run from his house to Tom's house. At 12:00 noon, both starts running toward each other's houses simultaneously. After they meet, both immediately turn around and run back to Jerry's house at Jerry's speed. How many minutes after noon will they arrive at Jerry's house?

【Answer】 24
【Solution】 Let the total distance be 1.
$1 \div (\frac{1}{21}+\frac{1}{28}) \times 2 = 24$ min

23. Find the smallest positive integer that has the sum of all of its prime factors equal to 2025.

【Answer】 30255
【Solution】 Smallest product = Largest difference
Hence, $2025 = 2017+5+3$
Our answer is just $2017 \times 5 \times 3 = 30255$

24. In a secondary school with 4 levels, the ratio of Sec 1 to Sec 2 students is 5:4, the ratio of Sec 2 to Sec 3 students is 7:8, and the ratio of Sec 3 to Sec 4 students is 9:7. If there are 2158 students in the school, how many Sec 1 student are there?

【Answer】 630
【Solution】 Since we know the ratio of them respectively, we can also combine all of the ratios together, to get the combined ratio of $S1:S2:S3:S4 = 315:252:288:224$
This would mean that $1058u = 2158$, $1u = 2$
There is a total of $315 \times 2 = 630$ students.

25. How many ways are there to divide 9 people into 3 groups of 3?

【Answer】 280
【Solution】 When we choose the people into groups, we use combination. But do note that we have to divide as the order does not matter within the group. ${}^9C_3 \times {}^6C_3 \times {}^3C_3 \div {}^3P_3 = 280$

26. 1
3 5
7 9 11
13 15 17 19
....
Find the number directly below 2025.

【Answer】 2115

【Solution】 2025 = 1013th position

We realize that every row is just +1, hence it is similar to an arithmetic sequence,

$1+2+3+4+...+n<1013$

Solving, when $n=44$, the sum is lesser than 1013.

Hence, 2025 is in Row 45.

the number directly below 2025 is (1013 + 45)th =1058 position.

$1058\times 2-1=2115$

27. The cost of 17 files, 25 pens, 19 erasers is \$25. The cost of 23 files, 21 pens and 31 erasers are \$36. What is the cost of 56 files, 108 pens and 52 erasers?

【Answer】 78

【Solution】 $\begin{matrix} 17x+25y+19z=25 \\ 23x+21y+31z=36 \end{matrix}$

$(17x+25y+19z)\times 6-(23x+21y+31z)\times 2=25\times 6-36\times 2$

$56x+108y+52z=78$

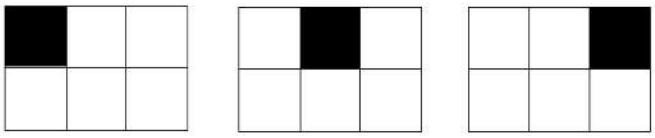
28. There are 6 identical square sheets of paper, with one face black and the other face white. We can put 6 square pieces of paper together to form a 2×3 rectangular cardboard. If two 2 × 3 cards cannot be rotated or flipped to make them look exactly the same, they are considered different. How many different 2 × 3 cards can be formed in total?

【Answer】 78

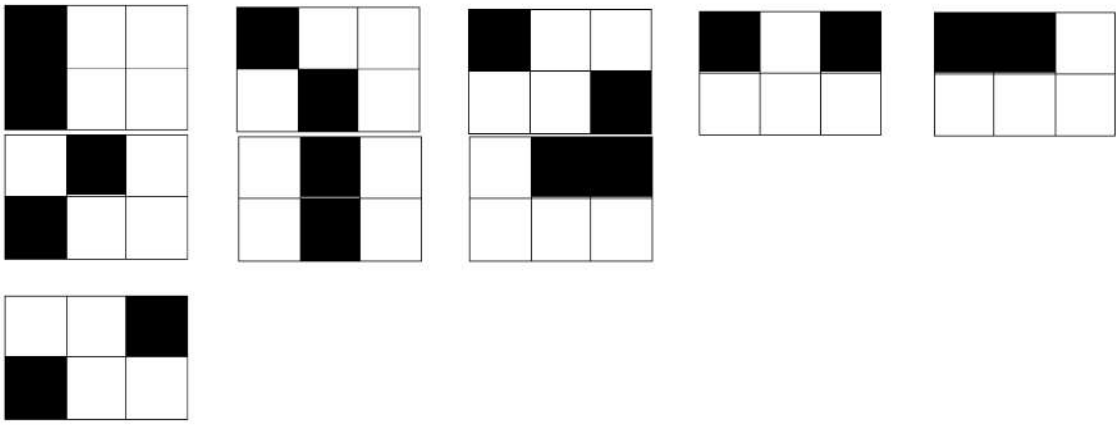
【Solution】

(a) 6 white/6 black: 1 way

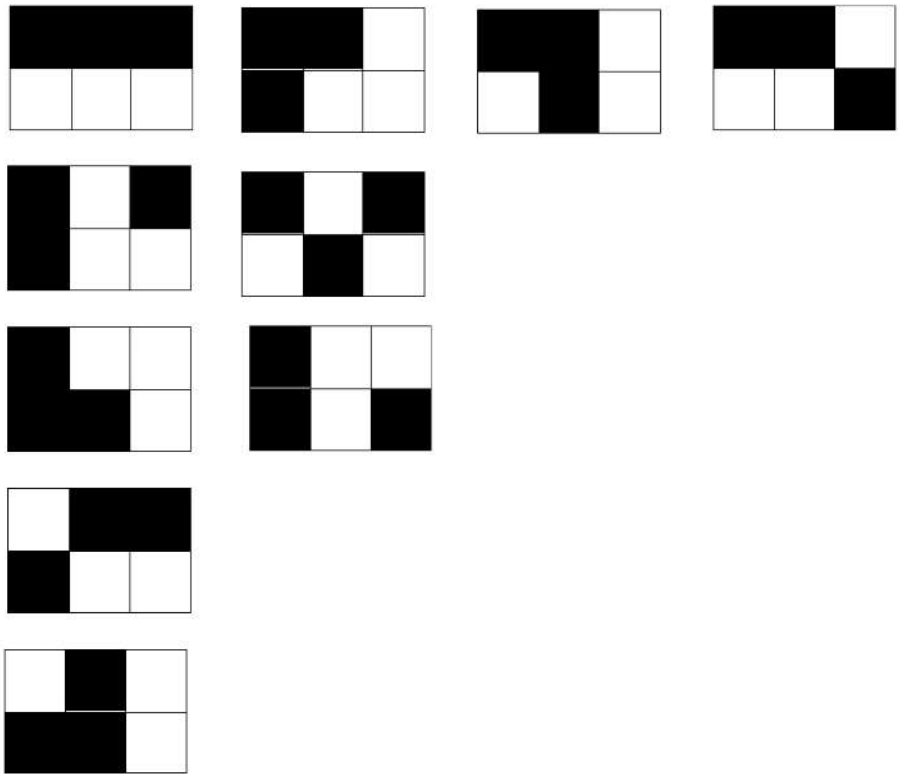
(b) 5 white and 1 black/1 white and 5 black: 3 ways



(c) 4 white and 2 black/2 white and 4 black:9 ways



(d) 3 white and 3 black: 10 ways



Total: 1 + 3 + 9 + 10 = 23 ways

29. There is a rectangle of area 1200, if its length increases by 10% and its breath decreases by 20%, what is the new area?

【Answer】 1056
【Solution】 $1200\times1.1\times0.8=1056$.

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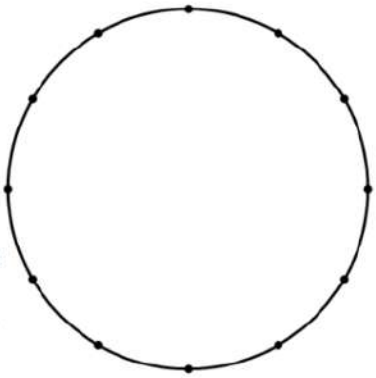
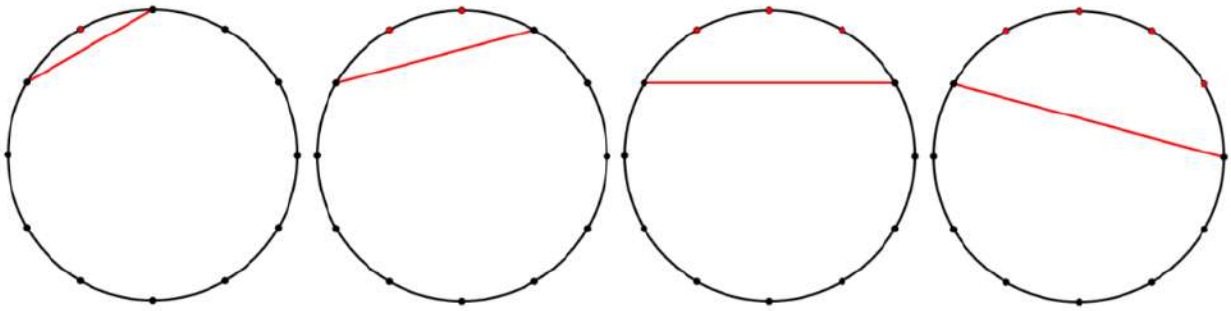
30. How many obtuse triangles can be formed by joining any of the 3 points?



【Answer】 78
【Solution】
As shown in the figure, 1 line segment and 1 point can form a triangle.
A line segments can divided the circle into 2 parts. To form an obtuse triangle, we need choose a point on the smaller part.
For a line segment with length "2" , "3", "4" and "5", there are 1, 2, 3 and 4 points respectively can be choose.
And each line segment with length "2" , "3", "4" and "5" has 12 choices.
So there are $(1+2+3+4)\times12=120$ obtuse triangle.